In a variant of hill-climbing, you allow sideways moves up to k steps. What issue does this address?

Global maxima
Plateaux
Loops
Random restarts

Given the heuristic function $h(n) = -(n-5)^2$, what value of n will a greedy hill-climbing algorithm converge to if starting from n=0 and allowed to increment by 1 each step?

0
4
5
10

Simulated annealing avoids getting trapped in local maxima by:

Random restarts
Periodic heuristic resets
Allowing downhill moves with a certain probability
Using deterministic search

Which of the following causes premature convergence in local beam search?

Using too many beams
Selection pressure
Low mutation rate
Lack of diversity in current states

Local beam search degenerates into hill-climbing when:

k = number of states
k = 1
$k \to \infty$
Random restarts are used

Assume a population of 100 individuals. After selection and crossover, only 5 unique individuals are produced. What does this imply?

High mutation rate
Optimal diversity
Unfit population
Premature convergence

You are given a genetic algorithm with crossover rate = 0.8 and mutation rate = 0.01. What happens if mutation is increased to 0.5?

Faster convergence
Loss of diversity
Increased randomness
Better fitness

In nondeterministic search, the solution must:

Guarantee success regardless of outcomes
Maximize expected utility
Always minimize cost
Be optimal for at least one outcome

Which strategy can handle the erratic behavior in the vacuum world?

AND–OR search
Depth-first search
A* search
Breadth-first search

What is the major challenge in searching under partial observability?

Lack of admissible heuristics
Inconsistent heuristics
Need to track belief states
Cost of sensors

Which of the following is an example of a partially observable game?

Chess
Poker
Checkers
Tic-Tac-Toe

Which of the following best describes the effect of good move ordering in alpha-beta pruning?

Reduces depth of search
Increases number of nodes evaluated
Maximizes pruning and improves efficiency
Has no effect on performance

In a stochastic game tree, what kind of node is introduced in addition to max and min nodes?

Leaf node
Utility node
Decision node
Chance node

Why can't standard alpha-beta pruning be applied directly to stochastic games?

They are not two-player
Their trees contain random chance nodes
They are partially observable
Alpha-beta pruning is only for infinite trees

Which game best exemplifies all three complexities: adversarial, stochastic, and partial observability?

Chess
Backgammon
Poker
Go